

Computational Astrophysics Exercises — Assignments Set 9

Classical hydrodynamical solver

In this exercise, you will implement a classical hydro solver following the ideas and instructions from the lecture on June 30th and July 7th. You can probably re-use some of your older scripts from the exercises, however, take your time to go through the single steps below.

(a) Consider the balance equations describing an inviscid, isothermal flow in one dimension

$$\begin{aligned}\partial_t Q_1 + \partial_x(Q_1 u) &= 0 \\ \partial_t Q_2 + \partial_x(Q_2 u) &= -\partial_x p\end{aligned}$$

with the vector $\mathbf{Q} = (\varrho, \varrho u)$, $u \equiv u_x$, with density ϱ , velocity in x -direction u and pressure p . Since we have an isothermal flow, we do not have to consider the balance equation for the energy. The pressure is given by the density times the constant speed of sound squared

$$p = c_s^2 \varrho.$$

Use the concept of operator splitting to handle the advection and the source term due to the pressure gradient separately.

The computational domain is $[0, 100]$ and the initial density profile is given by a Gaussian around the center (see fig. 1a)

$$\varrho = 1 + 0.3e^{-(x-50)^2/10}.$$

Integrate the equations from $t = 0$ to $t = 100$ with a time step that fulfills the Courant condition with Courant number 0.5 and reflective boundaries. The speed of sound is constant $c_s = 1.0$. Use a grid size of $N = 500$. Write an output file for each time step showing the density and produce a short video showing the evolution.

(b) Now, we consider also non-isothermal flows with adiabatic compression and expansion. We have to solve additionally the balance equation for the total energy

$$\partial_t(\varrho \epsilon_{\text{tot}}) + \partial_x(\varrho \epsilon_{\text{tot}} u) = -\partial_x(pu),$$

with $\epsilon_{\text{tot}} = u^2/2 + \epsilon$. The pressure is given now by the following equation of state

$$p = (\gamma - 1)\varrho\epsilon,$$

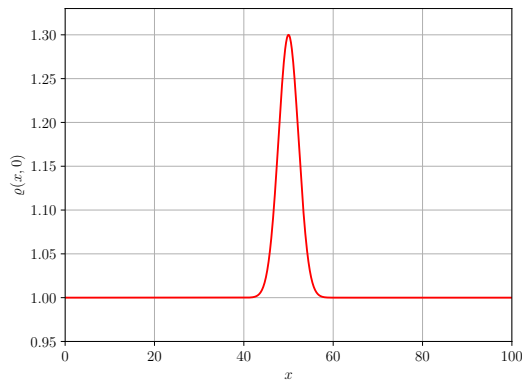
where γ denotes the adiabatic exponent. The sound speed is not constant anymore and given by

$$c_s = \sqrt{\gamma(\gamma - 1)\epsilon}.$$

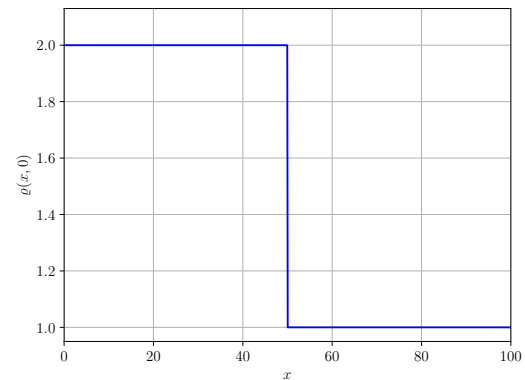
To test the code, we take a look at adiabatic shock. The initial condition is shown in fig. 1b. The density is $\varrho = 2$ for $x \leq 50$ and $\varrho = 1$ for $x > 50$, the initial internal energy is 1 and the velocity is $u = 0$. Integrate the equations from $t = 0$ until $t = 40$. Use $\gamma = 1.4$ for the equation of state. Note, that the sound speed is not constant anymore and you have to calculate the

time step size according to the current sound speed from the Courant condition for each time step. Create a movie showing the time evolution of the density, velocity and internal energy in the computational domain.

(c) Artificial viscosity is used to avoid oscillations near shock fronts in classical hydro codes. Add the von Neumann-Richtmyer artificial viscosity (with $\xi = 3.0$, see the lecture slide #28) to your program from exercise (b) and create another movie. Can you spot any difference?



(a) Exercise (a).



(b) Exercise (b).

Figure 1: Initial conditions for the test problems.

Hint for the movies: A nice and straightforward way to create a movie or animation of a sequence of images is by the help of the open-source software `ffmpeg`, see <https://ffmpeg.org/>.

points in total

(digits 49 and 50 of π , not counting 3.)

Deadline:

Fri, 10 July 2026 at noon  via email (including code) to `christoph.schaefer@uni-tuebingen.de`.